



THE DEPARTMENT OF CSE

PROJECT SYNOPSIS

Student Name: Abhishek Chatterjee, Radhey Mahato, Jiten Kr Sarkhel

Enrollment No.: SBU221026, SBU220349, SBU220845

Program & Branch: BTech CSE AI&ML

Batch: 2022-26, **Semester:** VII **Section:** A

Academic Session: Odd Semester 2025-26

Project Guide Details:

Guide Name: Dr. Dipti Kumari

Designation: Associate Professor, CSE, SBU

PROJECT INFORMATION

1. Course Title: Project Stage-II (Minor Project/Industrial Training)

2. Courses Code: CSE-702-P2

3. Credit Unit: 04

4. Project/ Internship Duration:

a) **Date of Project Commencement:** 25th September, 2025

b) **Synopsis Submission Date:** 6th October, 2025

c) **Date of Project Completion:** 22nd November, 2025

5. Approved Project Title (to be duly approved by the Project Guide):

“Plant Disease Detection and Classification Using Deep Learning”

6. Objectives:

- Develop a deep learning model to detect plant leaf diseases from images.
- Preprocess and train a CNN model using the Plant disease dataset.
- Build a Streamlit-based real-time disease prediction interface.
- Support early disease detection to improve crop productivity.
- Generate accurate class-wise predictions with performance statistics.

7. Methodology to be adopted:

- **Dataset Collection:** Use Plant disease images for training/testing.
- **Preprocessing:** Resize, normalize, augment, and split images.
- **Model Training:** Train a CNN/TensorFlow model to achieve high accuracy.
- **Evaluation:** Analyze accuracy, loss graphs, and confusion matrix.
- **Deployment:** Integrate trained model into a Streamlit web app.
- **Testing:** Validate predictions using unseen leaf images.
- **Documentation:** Prepare WPRs, report, and presentation.

8. Brief Summary of the project:

This project develops a deep learning-based plant disease detection system using CNN and the Plant disease dataset. The model learns to identify multiple crop diseases with high accuracy. A Streamlit web application is designed to allow users to upload leaf images and receive instant predictions with probability scores. The system helps farmers detect diseases early, supports better crop management, and promotes efficient, technology-driven agriculture.

Signature with date
(Student)

Signature of Industry Guide, if any

Signature with date
(Project Guide)